21 Show that $\dfrac{3+\sqrt{2}}{5+\sqrt{8}}$ can be written as $\dfrac{11-\sqrt{2}}{17}$

$\sqrt{8} = 2\sqrt{2}$. Multiply top and bottom by $5 - 2\sqrt{2}$.

$$\frac{3+\sqrt{2}}{5+2\sqrt{2}} \times \frac{5-2\sqrt{2}}{5-2\sqrt{2}}$$

1 mark
Conjugate factor

$$\frac{15-6\sqrt{2}+5\sqrt{2}-4}{25-10\sqrt{2}+10\sqrt{2}-8}$$

1 mark
Expand products

$$= \frac{11-\sqrt{2}}{17} \quad \text{as required.}$$

1 mark
Fully shown result

(Total for Question 21 is 3 marks)